



Department of Electronics and Communication Engineering
Academic Year 2021- 2022 (Odd Semester)

Degree, Semester & Branch: III Semester B.E. ECE-A

Course Code & Title: EC8352 Signals and Systems

Name of the Faculty member: Mrs.G.Gnana Priya

Innovative Practice Description

- **Unit / Topic:** Unit II / Laplace Transform – Properties
- **Course Outcome:** CO 2
- **Topic Learning Outcome:** TLO 7
- **Activity Chosen:** Flipped Classroom
- **Justification:**

The Unit II consists of Fourier transform and Laplace transform. The relation between Fourier transform and Laplace transform is very important. So I taught the properties of the Fourier transform in the class and then planned to conduct the properties of Laplace transform as a Flipped Classroom activity to make the students understand the two concepts clearly which enhances their learning level. From their presentation I can observe the understanding level of the students.

- **Time Allotted for the Activity:** 50 minutes
- **Details of the Implementation:**

The materials related to the activity were posted earlier in canvas for learning by the students. The activity was implemented by forming teams.

No. of teams: 10

No. of properties: 10

The team forming choice was given to the students with the condition that a team should consist of mixer of advanced learners, average learners and slow learners. In such a way the class strength was divided into 10 teams. Each team prepared and presented one property of Laplace transform.

CO – PO / PSO mapping:

CO	PO1	PO2	PO8	PO9	PO10	PSO1	PSO3
CO2	2	2	2	3	3	2	2

(1 – Low 2 – Moderate 3 – High)

• PO / PSO mapped:

Innovative practice	PO1	PO8	PO9	PO10	PSO1
	3	2	3	2	3
Justification for correlation	Analysis of CT signals requires the knowledge of mathematical and engineering fundamentals such as integration, differentiation, and transforms. So, the PO1 is highly correlated.	The PO8 is moderately correlated, since the Flipped Classroom activity involves presentation which applies moral & ethical principles while collecting the materials related to the topic.	Through this flipped classroom activity, the students work as a member of a team to achieve the goal. So, the PO9 is highly correlated.	Through flipped classroom activity the students' listen to, comprehend information, instructions and viewpoints of others. So, the PO10 is mapped at level 2.	The students will be able to innovate new ideas to solve existing/novel problems in the analysis of CT signals. So, the PSO1 is mapped at level 3.

• Images / Screenshot of the practice:



- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

The following points were observed based on the feedback got from the students.

- ✓ The students felt that the self-learning capability improved because of this activity.
- ✓ The presentation skills also increased while learning the concept by themselves and explaining to the class.
- ✓ The communication skills of the students improved while discussing with their team members and then presenting to the class.

- ❖ ***Benefit of the practice:***

- ✓ The main benefit of the flipped class activity is each group will learn one property and the presentation helps the other members also to know about all the properties.
- ✓ The students effectively involved in the activity by discussing with their team members.
- ✓ The understanding level and learning level of the students also increased.
- ✓ As the team forming choice was given to the students they actively participated.

- ❖ ***Challenges faced in implementation:***

- The biggest challenge was to making the students to learn the materials. Each student has to read the material posted in the canvas and then they have to prepare.
- Very few students were not actively participated in the activity. Making all the students to participate was also challenging.
- The activity was planned for one hour but it took two hours to complete as few teams took extra time than the allotted time to complete the activity.

References:

- ❖ Allan V.Oppenheim, S.Wilsky and S.H.Nawab, “Signals and Systems”, Pearson, 2015.
- ❖ <https://nptel.ac.in/courses/108/104/108104100/>



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Innovative Practice Description

- **Unit / Topic:** Unit III / Differential Equation- Laplace transforms in Analysis of CT Systems - Total Response

- **Course Outcome:** CO 3

- **Topic Learning Outcome:** TLO 11

- **Activity Chosen:** Round Table

- **Justification:**

Round Table activity is a method of round table discussion for solving problems which is used to engage students in ways that helps them integrate new and interesting content knowledge with prior knowledge through a structured round table format. I have chosen this topic Differential Equation- Laplace transforms in Analysis of CT Systems - Total Response for conducting this activity because the natural response and forced response were already taught to the students. So while finding the total response by round table discussion it enables them to solve the problems easily and effectively.

- **Time Allotted for the Activity:** 50 minutes

- **Details of the Implementation:**

- ❖ The students were asked to turn from their position of seating in such a way that they have to sit in groups (round table).
- ❖ Then each group was assigned a problem to solve
- ❖ The students referred the solved problems of natural response and forced response to find the total response
- ❖ They discussed with their group members to solve the problem
- ❖ The doubts were also clarified during the activity

CO – PO / PSO mapping:

CO	PO1	PO2	PO5	PO8	PO9	PO10	PSO1	PSO3
CO3	3	2	2	2	3	2	3	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO8	PO9	PO10	PSO1
	3	2	2	3	2	3
Justification for correlation	The students apply the knowledge of mathematics such as differential equation, Integration to solve problems in Linear Time Invariant Continuous Time Systems. So, the outcome is highly correlated.	The students analyze complex engineering problems in LTI CT systems and they will be able to find the response using mathematical principles. So, the outcome is mapped at level 2.	The PO8 is mapped at level 2 since the Collaborative learning round table activity is planned here so the students must follow basic ethical practice among themselves and within groups.	The PO9 outcome is highly correlated since the students work well as individuals and as a team while doing the round table activity.	The students' listen to, comprehend information, instructions and viewpoints of others through round table (collaborative) activity. As a result, PO10 is mapped to level 2.	The students come up with new ways to solve existing/new problems while determining the response to an LTI CT system. As a result, the PSO1 is mapped at level 3.

- **Images / Screenshot of the practice:**





Reflective Critique:

❖ *Feedback of practice from students and other stakeholders:*

The following points were observed based on the feedback got from the students.

- ✓ The slow learners felt this activity very useful for solving problems by discussing with their friends who know the concept very clearly.
- ✓ By sharing the steps involved in solving the problem with their group members it enhances their communication skills.

❖ *Benefit of the practice:*

- ✓ The main benefit of this collaborative activity is to make the slow learners to complete the task with the support of their group members.
- ✓ The members of the team verified the answers of each step before executing the next step, which helped them to avoid mistakes.
- ✓ They also discussed effectively within the team.
- ✓ This type of team formation helped the slow learners to solve the problem in time.

❖ *Challenges faced in implementation:*

- In few groups the interest in doing collaborative learning was missing.

References:

- ❖ Allan V.Oppenheim, S.Wilsky and S.H.Nawab, “Signals and Systems”, Pearson, 2015.
- ❖ <https://nptel.ac.in/courses/108/104/108104100/>



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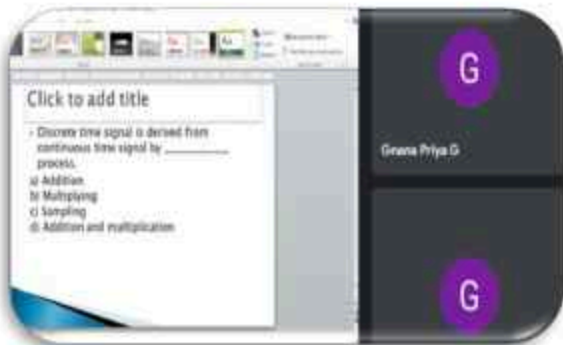
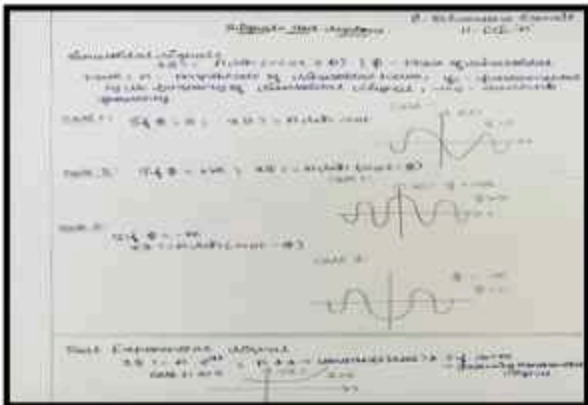
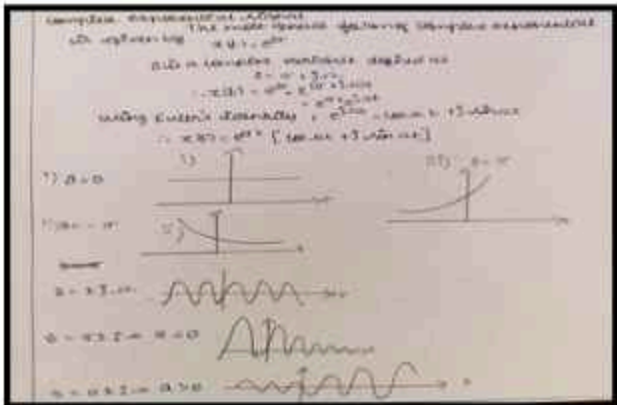
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INNOVATIVE TEACHING METHOD

Degree, Semester & Branch: III Semester B.E. ECE A

Course Code & Title: EC8352 Signals and Systems

Name of the Faculty member: Mrs.G.Gnana Priya

Sl.No.	Topic(s)	Activity	Reference(s)
UNIT I - CLASSIFICATION OF SIGNALS AND SYSTEMS			
1.	Introduction- Standard signals- Step, Ramp, Pulse	Quesscussion	Allan V. Oppenheim, S. Willsky and S. H. Nawab, "Signals and Systems", Pearson Education, 2007.
 			
2.	Standard signals-Real and complex exponentials and Sinusoids	Minute Paper	Allan V. Oppenheim, S. Willsky and S. H. Nawab, "Signals and Systems", Pearson Education, 2007.
 			

3.	Classification of systems- Stable & Unstable, Revision	Crowd Sourcing	https://nptel.ac.in/courses/108/104/108104100/
			



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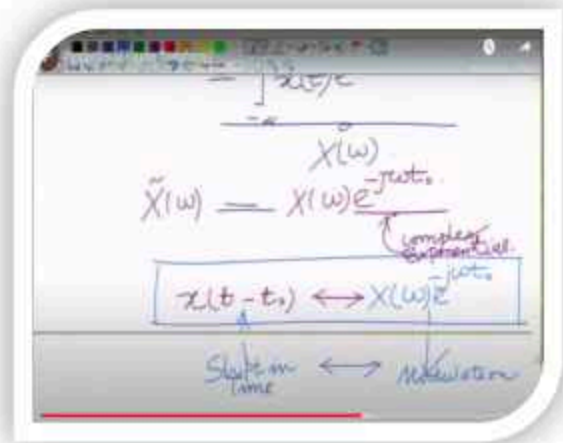
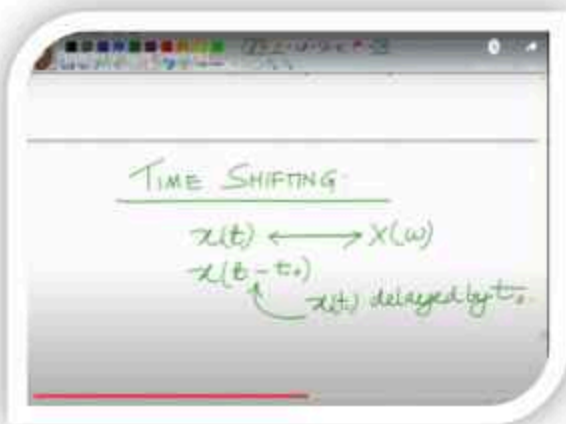
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Sl.No.	Topic(s)	Activity	Reference(s)
UNIT II - ANALYSIS OF CONTINUOUS TIME SIGNALS			
1.	Fourier series for periodic signals – Exponential Fourier series	Clickers	Allan V. Oppenheim, S. Willsky and S. H. Nawab, "Signals and Systems", Pearson Education, 2007.
 			
2.	Fourier series for periodic signals – Properties of Fourier Series -Linearity, Time shifting, Frequency Shift, Time scaling, Time Differentiation, Convolution, Multiplication, Parseval's theorem	Open book test (library)	Allan V. Oppenheim, S. Willsky and S. H. Nawab, "Signals and Systems", Pearson Education, 2007.
 			

3.

Fourier Transform - Properties

NPTEL video

<https://nptel.ac.in/courses/108/104/108104100/>



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Sl.No.	Topic(s)	Activity	Reference(s)
UNIT III - LINEAR TIME INVARIANT –CONTINUOUS TIME SYSTEMS			
1.	Convolution integrals - Properties	Head and Talk	Allan V. Oppenheim, S. Willsky and S. H. Nawab, "Signals and Systems", Pearson Education, 2007.
 			
2.	Systems connected in series / parallel, Revision	Memory Matrix	Allan V. Oppenheim, S. Willsky and S. H. Nawab, "Signals and Systems", Pearson Education, 2007.
 			

3.	Convolution integrals	Online Simulation	https://www.youtube.com/watch?v=LZ0qjZezGkQ
			



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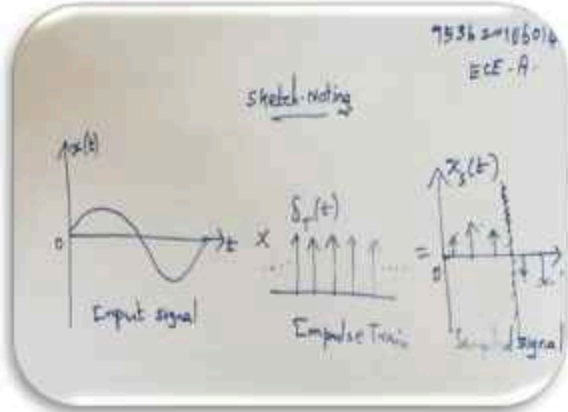
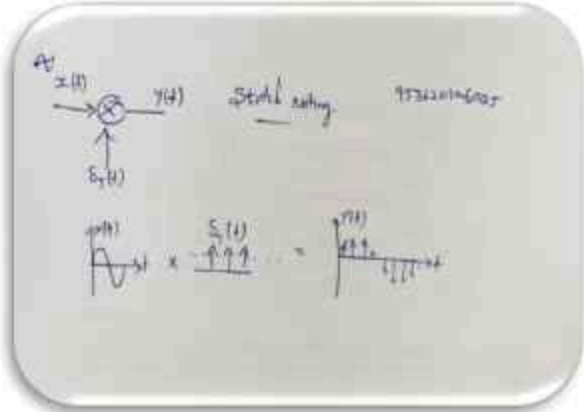
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Sl.No.	Topic(s)	Activity	Reference(s)
UNIT IV - ANALYSIS OF DISCRETE TIME SIGNALS			
1.	Baseband signal Sampling	Sketch Noting	Allan V. Oppenheim, S. Willsky and S. H. Nawab, "Signals and Systems", Pearson Education, 2007.
 			
2.	Properties of Z-transform	Index Card Match	Allan V. Oppenheim, S. Willsky and S. H. Nawab, "Signals and Systems", Pearson Education, 2007.
 			

3.	Methods of Inverse Z Transform	Mind Map	https://www.youtube.com/watch?v=LZ0qjZezGkQ
			



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Sl.No.	Topic(s)	Activity	Reference(s)
UNIT V - LINEAR TIME INVARIANT - DISCRETE TIME SYSTEMS			
1.	Convolution sum	Sorting strips	Allan V. Oppenheim, S. Willsky and S. H. Nawab, "Signals and Systems", Pearson Education, 2007.
 			
2.	Z Transform in analysis of Recursive & Non-Recursive systems	Finger Signal	Allan V. Oppenheim, S. Willsky and S. H. Nawab, "Signals and Systems", Pearson Education, 2007.
 			

3.	DT systems connected in series and parallel	Zero Minute Speech	https://www.youtube.com/watch?v=MVygzbbUDc
			
